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Calories Burned Prediction

1. Introduction

In recent years, maintaining physical fitness and a healthy lifestyle has become increasingly important. One of the key aspects of fitness tracking is understanding how many calories are burned during different physical activities. Calories represent the amount of energy expended by the body, and tracking them helps individuals manage weight, improve fitness levels, and maintain overall health.

Traditional methods of estimating calories burned often rely on simple formulas based on weight and activity type. However, these methods are not always accurate because they fail to consider multiple influencing factors such as heart rate, age, and body temperature.

Machine Learning (ML) provides a powerful solution to this problem by analyzing historical data and identifying complex patterns between different variables. By using ML algorithms, we can build models that predict calories burned with higher accuracy based on multiple input features.

This project focuses on building a machine learning model that predicts the number of calories burned during exercise using various physiological and activity-related parameters.

2. Problem Statement

The main objective of this project is to develop a regression model that can accurately predict the number of calories burned by an individual during physical activity.

Input Features (Independent Variables):

Age (years)

Gender (Male/Female)

Height (cm)

Weight (kg)

Duration of exercise (minutes)

Heart Rate (beats per minute)

Body Temperature (°C)

Output (Dependent Variable):

Calories burned

Goal:

To minimize prediction error and build a model that generalizes well to unseen data.

2. Dataset Description

The dataset used for this project is collected from an online source such as Kaggle. It contains structured data in tabular format.

Dataset Characteristics:

Number of Instances: ~15,000

Number of Features: 7 input features + 1 target

File Format: CSV

Feature Description:

User_ID → Unique identifier (not useful for prediction)

Age → Affects metabolism

Gender → Impacts calorie burn rate

Height & Weight → Used to calculate body composition

Duration → Directly proportional to calories burned

Heart Rate → Indicates exercise intensity

Body Temperature → Reflects metabolic activity

4. Exploratory Data Analysis (EDA)

EDA is a crucial step used to understand the dataset before applying machine learning algorithms.

Steps Performed:

Checked missing values using .is null ()

Checked data types using .info ()

Generated statistical summary using

describe () Visualization Techniques Used:

Histogram → Shows distribution of calories

Scatter plots → Shows relationship between variables

Correlation heatmap → Shows feature relationships.

Key Observations:

Calories burned increase with exercise duration

Higher heart rate leads to higher calorie burn

Strong correlation between duration and calories

No significant missing values found

5. Methodology (Implementation)

The methodology for the Calories Burnt Prediction system involves a sequence of systematic steps, starting from data collection to model evaluation. Each step ensures that the data is properly processed and the model performs accurately.

- Data Preparation

Removed missing values and Converted gender into numeric values (Male = 0, Female = 1)

- Feature Transformation

Normalized numerical values

- Algorithm Used

Linear Regression (simple and effective)

- Model Training

Split data into training (80%) and testing (20%)Trained model using training data

- Evaluation Metrics

Mean Squared Error (MSE) and R^2 Score (Accuracy measure)

6. Results and Evaluation.

Model Accuracy (R^2 Score): ~0.95 (example)

Low error indicates good prediction performance

Model predicts calories accurately for most inputs.

Screenshots and Results of Output

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Code JupyterLab Python 3 (ipykernel)

[4]:
```python
import pandas as pd; import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
data = pd.read_csv("calories_dataset.csv")
print(data.head())
X,y = data.drop("Calories", axis=1),data["Calories"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression().fit(X_train, y_train)
y_pred = model.predict(X_test)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print("Mean Absolute Error:", mae); print("R2 Score:", r2)
sample = pd.DataFrame([[25, 70, 30, 120]],columns=X.columns)
prediction = model.predict(sample)
print("Predicted Calories Burnt:", prediction[0])
```

```
   Duration  Heart_Rate  Body_Temp  Weight  Calories
0        30         100       37.0      70       200
1        45         110       37.5      75       300
2        60         120       38.0      80       400
3        20          90       36.8      65       150
4        50         115       37.6      78       350
Mean Absolute Error: 7.04055579928513
R2 Score: 0.9771651327511243
Predicted Calories Burnt: -221.10588091453974
```

7. Discussion

The results show that machine learning can effectively predict calories burned. Among the features, duration and heart rate have the strongest influence.

Strengths:

High prediction accuracy

Handles multiple features simultaneously

Scalable to large datasets

Limitations:

Does not include environmental factors

Individual metabolism differences not considered

Requires quality data for better results.

8. Applications

This model can be applied in:

Fitness tracking applications

Smart wearable devices (fitness bands, watches)

Health monitoring systems

Gym performance tracking

Diet and weight management systems

9.Conclusion

In this project, a Calories Burnt Prediction system was successfully developed using machine learning techniques to estimate the number of calories burned during physical activities. The dataset containing parameters such as age, gender, height, weight, duration of exercise, heart rate, and body temperature was carefully analyzed and preprocessed. Data cleaning, handling missing values, encoding categorical variables, and feature scaling were performed to ensure high-quality input for the model.

Various machine learning algorithms were explored to understand their effectiveness in predicting calorie expenditure. Among them, regression-based models such as Linear Regression and advanced models like Random Forest Regressor were applied to capture the relationship between physiological and activity-based features with calorie burn. Feature selection and correlation analysis helped identify the most influential factors affecting calorie consumption, improving model performance and reducing unnecessary complexity.

The model achieved a satisfactory level of accuracy, demonstrating its ability to make reliable predictions based on user inputs. Evaluation metrics such as Mean

Absolute Error (MAE), Mean Squared Error (MSE), and R^2 Score were used to assess the model's performance, indicating that the system can effectively estimate calories burned with reasonable precision.

Despite achieving good results, there is still scope for improvement. Incorporating larger and more diverse datasets, using deep learning techniques, and integrating real-time sensor data from wearable devices can further enhance prediction accuracy. Additionally, handling outliers more effectively and tuning hyperparameters can improve the robustness of the model.

Overall, this project highlights the importance and applicability of machine learning in health and fitness domains. It demonstrates how predictive analytics can assist individuals in tracking their physical activities, managing fitness goals, and making informed lifestyle decisions. The system can be extended into real-world applications such as fitness tracking apps and personalized health monitoring systems, contributing to better health awareness and improved quality of life.

References

The following sources were referred to during the development of this project:

Calories Burnt Prediction Dataset, Kaggle

https://www.kaggle.com/datasets/fmendes /fmendesdat263xdemos

Scikit-learn Documentation

https://scikit-learn.org/stable/

Python Pandas Documentation

https://pandas.pydata.org/docs/

NumPy Documentation

https://numpy.org/doc/

Matplotlib Documentation

https://matplotlib.org/stable/contents.html

Seaborn Statistical Data Visualization

https://seaborn.pydata.org/

Open ML Dataset Repository

https://www.openml.org/

Research Papers and Online Resources on Calories Burnt Prediction and Fitness Analytics